Individual Contribution Journal: Lab 07 Model Evaluation

Course: ITAI-1371 – Fundamentals of Artificial Intelligence

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Assignment: Lab 07 Model evaluation

1. **Project Management, Repository Architecture & Group Coordination**

- GitHub Repository Setup & Administration: I took the initiative to set up and structure our official group GitHub repository. I organized the folder layout for our Colab notebooks, dataset references, exported figures, and markdown documentation so our version control was structured and reproducible.
- Team Follow-Up & Workflow Coordination: I served as the primary coordinator for our team milestones, following up with teammates via our group channel, tracking completion of individual tasks, and ensuring everyone was aligned on the submission deadline.
- Document Compilation & Final Deliverable Assembly: I collected all team members' components, collated our notes, verified that the executed notebook and plots were intact, and assembled the final submission documents into standardized PDF formats adhering to the course naming requirements (L07_..., L07Journal_R_..., and L07Journal_C_...).

2. Technical Implementation & Pipeline Debugging (Google Colab)

- Data Cleaning & Pipeline Debugging (Part 1): I resolved syntax errors in the initial starter code caused by merged code lines. I standardized the preprocessing steps by mapping categorical strings (Sex to binary 0 and 1), imputing missing passenger ages using median replacement (df['Age'].fillna(df['Age'].median())) to eliminate deprecation warnings, and training our LogisticRegression(max_iter=1000) model on an 80/20 split (random_state=42).
- Confusion Matrix Visualization (Task 1): I implemented confusion_matrix(y_test, y_pred) and used Seaborn to generate an annotated heatmap (sns.heatmap) with direct integer formatting (fmt='d') and categorical class labels (Died vs. Survived) to visualize Type I and Type II errors.
- Classification Report Analysis (Task 2): I generated the full classification report using sklearn.metrics.classification_report. I evaluated the trade-offs between our majority class (Died: 0.82 precision, 0.86 recall) and minority class (Survived: 0.78 precision, 0.73 recall), pointing out to the group that our 80% baseline accuracy hid a higher rate of False Negatives for passenger survival.

- Cross-Validation Implementation (Task 3): I constructed the 5-fold cross-validation routine using cross_val_score(model, X, y, cv=5). I extracted the scores across all five folds ([0.7877, 0.7809, 0.7865, 0.7753, 0.8034]), computed our mean generalization performance (78.68%), and calculated the standard deviation ($\sigma = 0.0094$) to demonstrate model stability across data partitions.

3. Knowledge Check Technical Formulation

- Precision Formulation: I defined precision ($\frac{\text{TP}}{\text{TP+FP}}$) in the context of high-cost Type I errors (such as automated banking fraud locks or false-positive email filtering), justifying why the decision threshold must be tuned to minimize False Positives.
- Recall Formulation: I defined recall ($\frac{\text{TP}}{\text{TP+FN}}$) in the context of high-cost Type II errors (such as oncological screening), justifying why diagnostic algorithms must prioritize sensitivity to avoid catastrophic False Negatives.
- Cross-Validation Statistical Defense: I wrote the technical explanation for why k-fold cross-validation ($\hat{\theta}_{\text{CV}} = \frac{1}{k}\sum_{i=1}^{k}\hat{\theta}_i$) reduces estimator variance ($\text{Var}(\bar{X}) = \frac{\sigma^2}{k}$) and prevents sampling bias compared to a single train-test split.

Module 07 Lab – Contribution Journal

ITAI-1371: Introduction to Machine Learning | Model Evaluation

Hamza Ajmal

Notebook setup and data flow

I reviewed the notebook setup and checked how the data, target labels, and train/test split were prepared for model evaluation. I verified that the evaluation steps used the intended test data and that the notebook cells were run in the correct order.

Confusion matrix

I worked on the confusion-matrix task and reviewed how the matrix separates correct predictions from false positives and false negatives. I checked the displayed matrix or heatmap and considered what the different error types reveal about the classifier.

Classification metrics

I completed or reviewed the classification report showing precision, recall, F1-score, and accuracy. I considered why the metrics describe different aspects of performance and why accuracy alone may not fully explain a model's behavior.

Cross-validation

I completed or reviewed the five-fold cross-validation task, checked the score for each fold, and considered the average score. This helped me compare repeated evaluation results rather than relying only on one train/test split.

Verification and troubleshooting

I checked the outputs for errors and confirmed that the notebook produced the expected evaluation results.